

RAK12044 WisBlock Hall Effect Sensor Datasheet

Overview

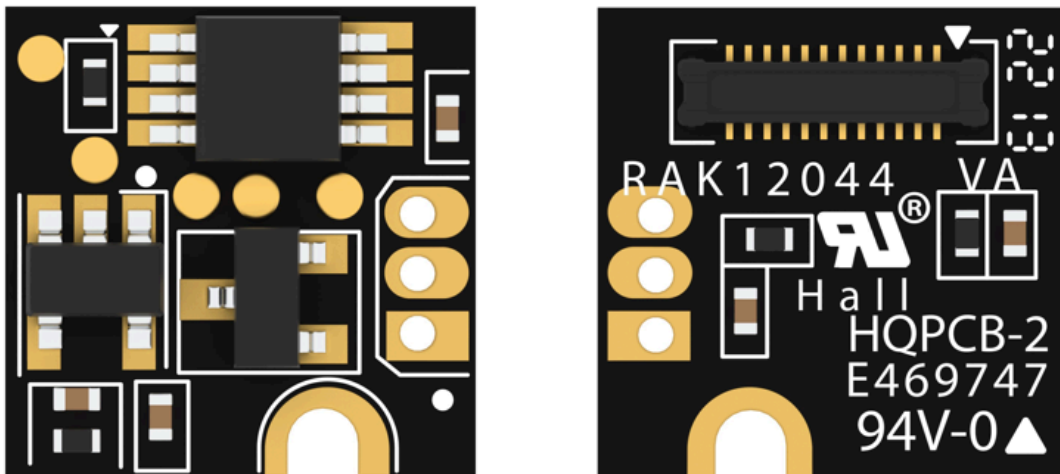


Figure 1: RAK12044 WisBlock Hall Effect Sensor

Description

RAK12044 is a WisBlock Hall Effect Sensor module that extends the WisBlock system with a DRV5056A4QDBZR Hall Sensor from Texas Instruments. It comes with a ready-to-use software library and tutorial, making it easy to build a magnetic field data acquisition system. Magnetic field data is interfaced via I2C. Additionally, the RAK12044 can be mounted to the sensor slot of the WisBlock Base board.

Product Features

- Sensor specifications
 - Based on DRV5056A4QDBZR
 - Magnetic sensitivity options: 15 mV/mT, 155-mT range

- I2C digital interfaces
- 3.3 V power supply
- Operating Temperature Range: -40 °C to +80 °C
- Size
 - 10 × 10 mm

Specifications

Overview

Mounting

Figure 2 shows the mounting mechanism of the RAK12044 module on a [WisBlock Base](#) board. The RAK12044 module can be mounted on the slots: A, C, D, E, & F.

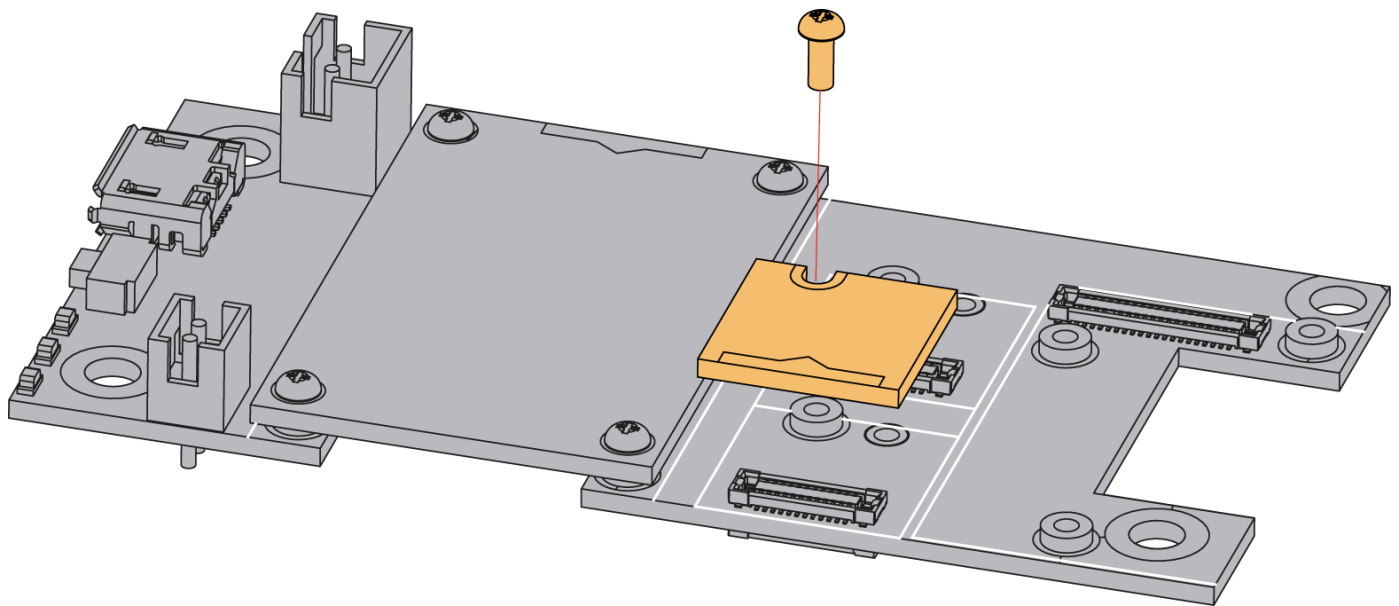


Figure 2: RAK12044 WisBlock Hall Effect Sensor Mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the

electrical and mechanical parameters, including the tabular data of the functionalities and standard values of the RAK12044 WisBlock Hall Effect Sensor.

Chipset

Vendor	Part Number
Texas Instruments	DRV5056A4QDBZR

Pin Definition

The RAK12044 WisBlock Hall Effect Sensor comprises a standard WisBlock connector. The WisBlock connector allows the RAK12044 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition are shown in **Figure 3**.

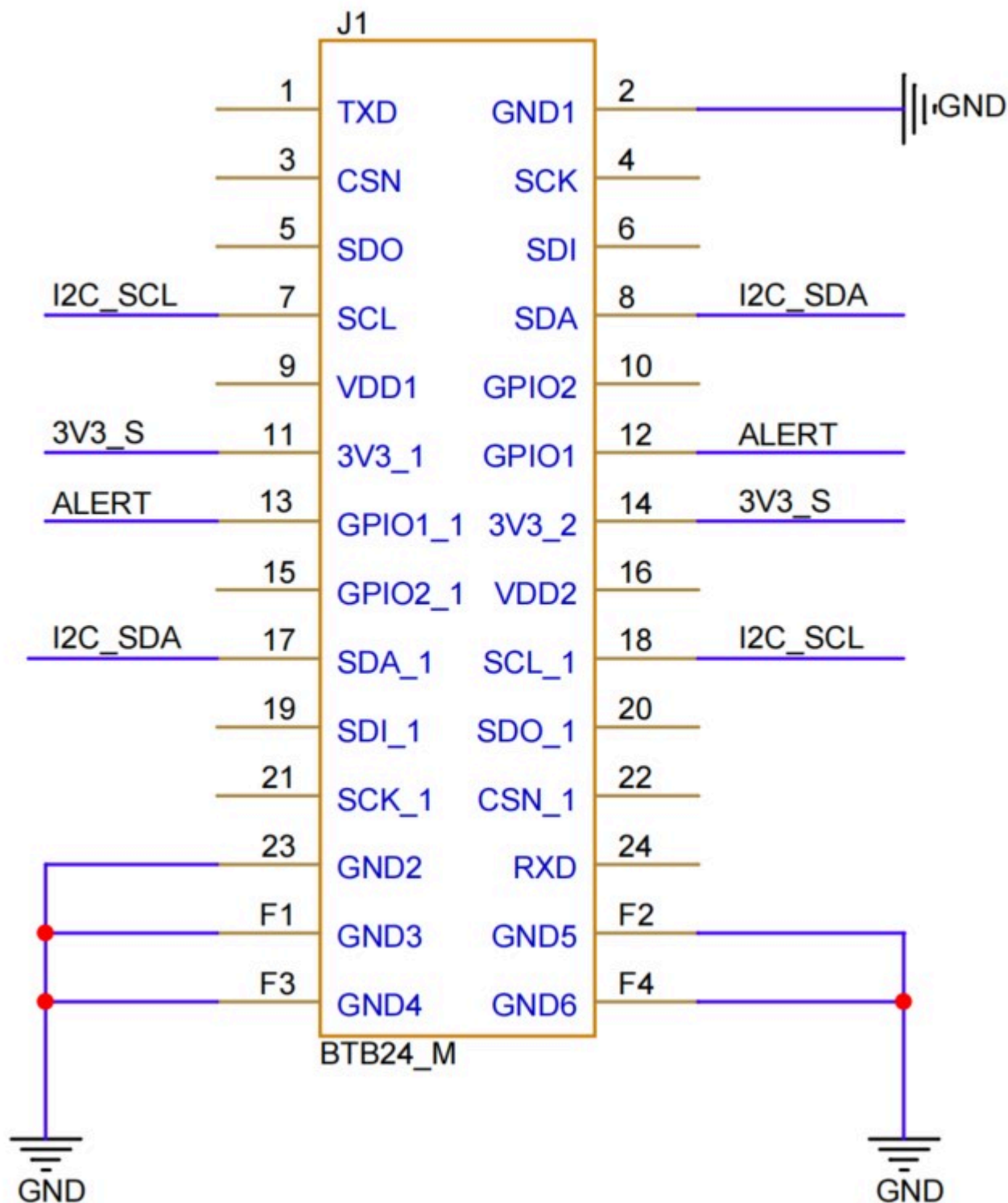


Figure 3: RAK12044 WisBlock Hall Effect Sensor Pinout Diagram

NOTE

When the RAK12044 is used with other sensor modules powered by **3V3_S** together, it can not be mounted on slot **B**.

The **WisBlock Sensor** connector is used to this module and the IO used for **ALERT** pin at **Pin 12** will depend on where sensor slot the module is plugged in. The table shows the compatible pins used by different sensor slots:

ALERT Pin

Slot A	Slot C	Slot D	Slot E	Slot F
IO1	IO3	IO5	IO4	IO6

Electrical Characteristics

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
3V3_S	Power Supply Voltage	-	3.3	-	V

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK12044 module.

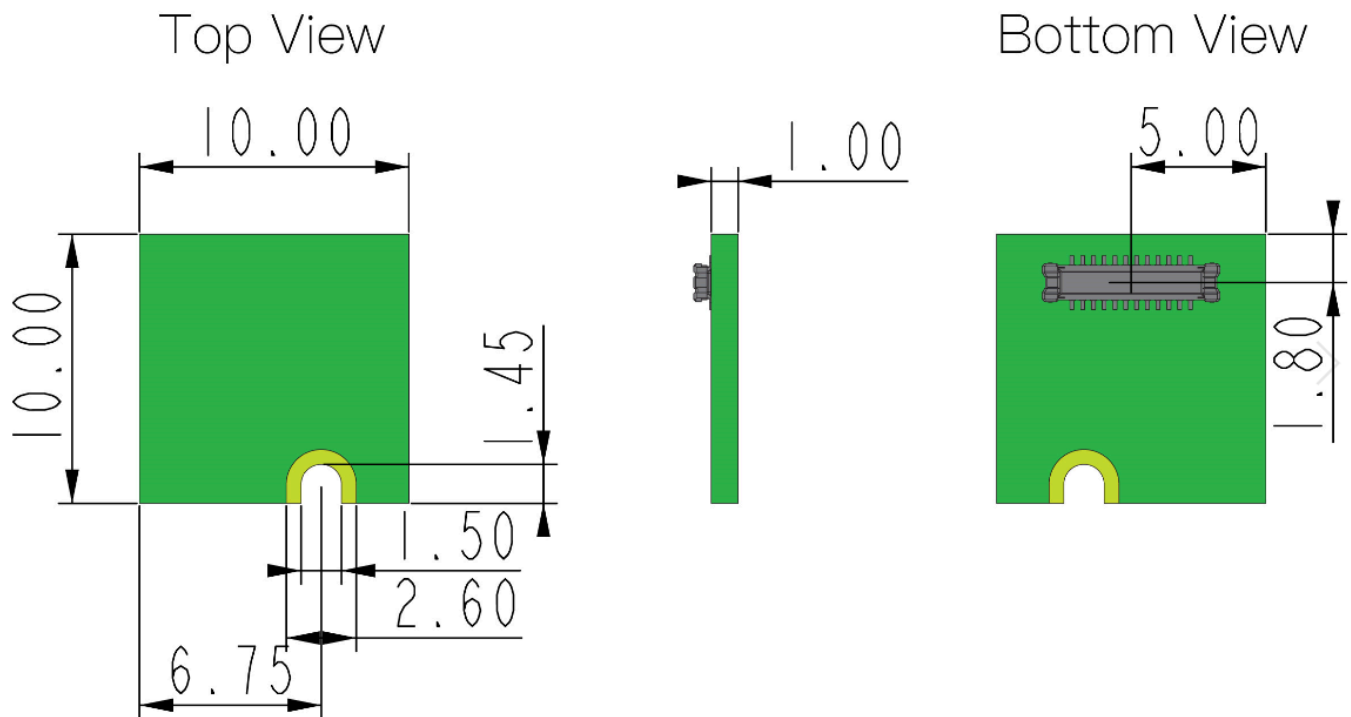


Figure 4: RAK12044 WisBlock Hall Effect Sensor Mechanic Drawing

WisConnector PCB Layout

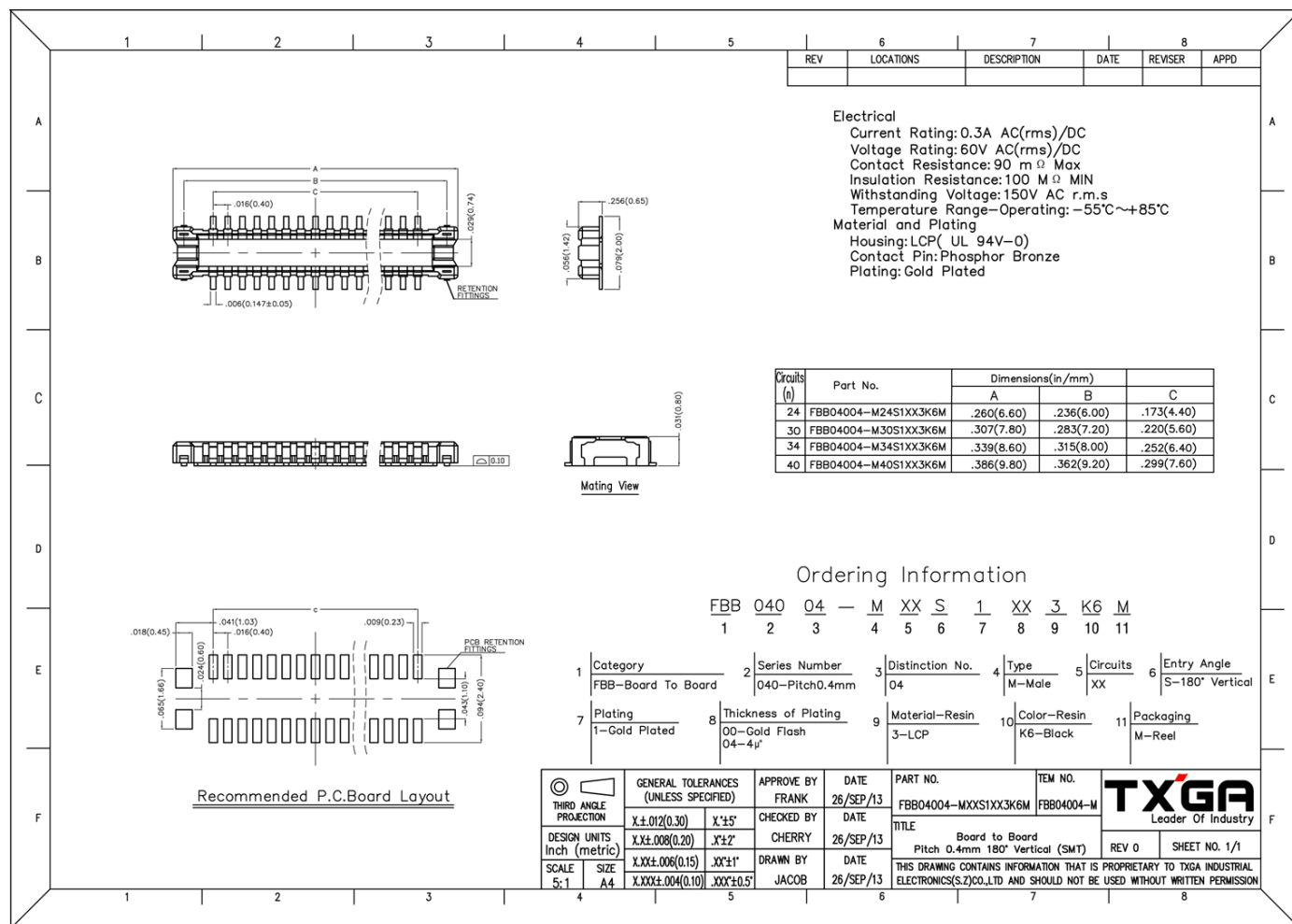


Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK12044 module.

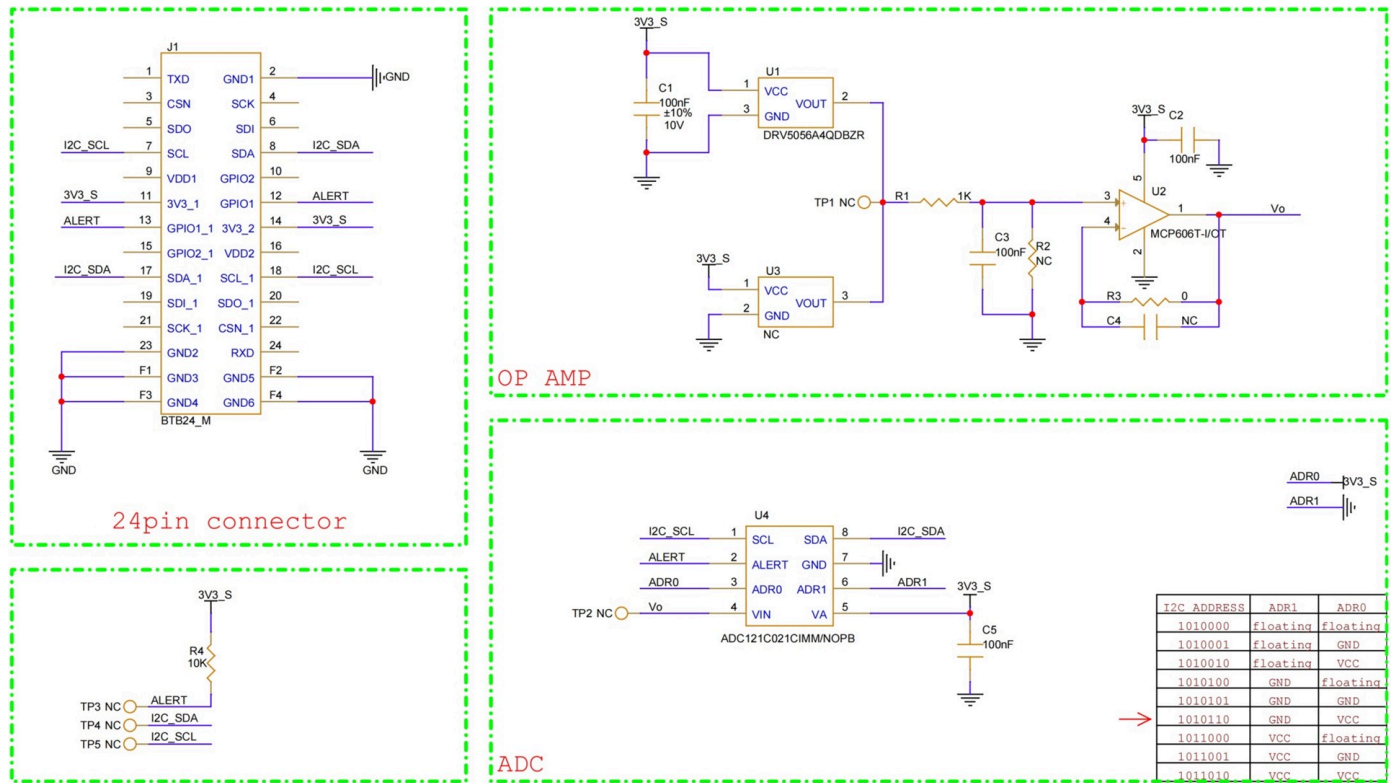


Figure 6: RAK12044 WisBlock Hall Effect Sensor schematics